

Switching Circuits Laboratory (CS29002)

Assignment 6 (11-03-2026)

- a) Design a J-K master slave flip-flop using NAND gates and verify the functionality. Hence, convert the J-K flip-flop to a T flip-flop and demonstrate that it divides the frequency of an incoming rectangular waveform by 2.
- b) Design a J-K latch using NAND gates. Connect an edge-detection circuit block that generates a very narrow pulse on every rising clock edge to drive the latch's Enable input. Hence, verify whether the edge-detection operation works correctly.

Instructions:

- 1) *Draw the circuit diagram on paper, following the suggested conventions. Following this, you can issue the components and start with the realization on the breadboard.*
- 2) *Prepare a laboratory report in PDF format and upload it to Moodle by the specified deadline. A single report has to be uploaded per group.*